NUS National University of Singapore	Department of Biological Sciences National University of Singapore	Ref. No	REL/SOP/21
Standard Operating Procedure **Title: Double-digest RAD sequencing**		**Rev. No**	1
		Pages	9
Lab: Reef Ecology Laboratory			

Written by	Approved by	Issue date	Review date
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1. Purpose

The objective of this SOP is to guide all laboratory personnel on performing double-digest restriction site-associated DNA sequencing (ddRADseq) sequencing for organism of interest.

2. Scope

The procedure is applicable to all research staff, research students and technical staff working in the laboratory.

3. Responsibility

It is the responsibility of the Principal Investigator (PI) to ensure that all research and technical staff and students are adequately advised, prepared and trained. All laboratory personnel must receive training prior to performing this procedure. Training may be delegated to a qualified individual in the laboratory but it remains the responsibility of the PI to ensure that laboratory personnel are adequately trained.

4. Preparative Work

4.1. <u>Adapters (from Peterson et al., 2012)</u>

- Resuspend lyophilized oligos in 1× Duplex Buffer to make 200µM stock:

$$Amount\ of\ Duplex\ Buffer\ (\mu l) = \frac{Amount\ of\ oligo\ (nmol) \times 10}{2}$$

- Anneal the P1.1E + P1.2E and P2.1M + P2.2M oligos to make P1E and P2M adapters respectively:

We want **5µM P1 adapters** (*EcoRI* cut site):
200µM stock P1.1E 5µl
200µM stock P1.2E 5µl
Duplex Buffer 190µl
Total = 200µl of 5µM working stock

We want **50µM P2 adapters** (*MspI* cut site):
200µM stock P2.1M 25µl
200µM stock P2.2M 25µl
Duplex Buffer 50µl
Total = 100µl of 50µM working stock

- Heat oligo mix to 97.5ºC for 2 mins 30 secs
- Reduce temperature by 2ºC per minute until 15ºC is reached
- Hold at 15ºC
- Store at -30ºC

4.2. SeraMag SpeedBeads in PEG (for size selection)

	Stock Concentration	*Dilution Factor*	*Amount needed for 25ml buffer*	*Amount needed for 500ml buffer*	*Final concentration*
PEG-8000	Powder	-	4.5g	90g	18% w/v
NaCl	Powder	-	1.461g	29.22g	1M
Tris-HCl pH 8	1M	100×	250µl	5ml	10mM
EDTA	0.1M	100×	250µl	5ml	1mM
Water	-	-	Top up to 25ml	Top up to 500ml	-

- To make 25ml 0.1M EDTA (MW 372.24), dissolve 0.9306 EDTA salt in 25ml of water

- SeraMag SpeedBeads in PEG preparation:
  - Wash carboxyl-modified SeraMag SpeedBeads (GE Healthcare 65152105050250) twice with equal volumes of sterile water, and once with PEG buffer (*beads must be washed prior use because original buffer contains sodium azide*)
  - Re-suspend beads with 50× volume buffer to initial volume of beads taken
  - Store at 4ºC till use
  - Test re-suspended beads using non-dyed DNA ladder for size selection range

5. Procedure

5.1. DNA QC and Normalisation

To check for gDNA quality and obtain roughly equal amounts of DNA per reaction

To prepare: Autoclaved plates [Set A**] Molecular-grade water Filtered tips

- Concentrate gDNA volumes to ~30µl
- Run 1–2µl of gDNA on 1% agarose gel to check for quality:
  - Genomic bands = sample is ideal for ddRADseq
  - Smearing = gDNA degradation, not ideal but worth a try if samples are limited

- Select some samples from each gel (based on gel band intensity) for Qubit and approximate for the rest:
  - Starting sample concentration must be ≥ 5.7ng/µl
  - Assign higher and lower concentrations to separate indexes
- Prepare 100–500ng of DNA in plates [**Set A**]:
  - Final volume ~21µl (gDNA + water) per sample
  - Increase accordingly if you have spare DNA and wish to repeat reactions

5.2. RELIG (RE Digest + Adapter Ligation)

Simultaneous double digest of gDNA and ligation of the adapters to sticky ends [5 hrs 30 mins]

**To prepare:
Autoclaved plates [**Set B**]
Silicone sealing mats or adhesive seals
Filtered tips
Tube for master mix
Ice box with ice

- Prepare master mix in the tube:

Reagent	*Final Concentration*	*1×*	
10× T4 DNA Ligase Buffer	1×	3.2µl	Master Mix [**Set B**]
0.5M NaCl	50mM	3µl	
20mg/ml BSA	0.05mg/ml	0.08µl	
50µM P2M adapter	3.85µM	2.31µl	
EcoRI-HF (20U/µl)	11.5U	0.58µl	
MspI (20U/µl)	2.3U	0.12µl	
T4 DNA Ligase (400U/µl)	185U	0.46µl	
5µM P1E adapter	0.385µM	2.31µl	**Set A**
gDNA + Water	-	17.94µl	
TOTAL	-	30µl	

- Aliquot 9.7µl of master mix into each well [**Set B**]
- Mix well and transfer 2.31µl of P1E adapters into **Set B**
- Mix well and transfer 18µl of **Set A** into **Set B**
- RELIG Conditions:
  - 37ºC for 4 hrs
  - 65ºC for 20 mins (heatkill step)
  - Hold at 4ºC
- NOTE: MspI is not inactivated by heat, preferable to clean-up products following RELIG

- OPTIONAL: Run ~1.5µl of selected samples on 1–2% agarose gel to check RELIG success, and to see if there is an optimal range to select size fragments from
    - Should observe smearing on the gel, along with absence of genomic band

5.3. SPRI Bead Clean-up and Size Selection

Simultaneous clean-up and one-ended size selection of RELIG products. For size selection, factor in desired size fragment (~400bp) along with adapter lengths (76bp) [1 hr – 1hr 30 mins]

To prepare: 1.5ml tubes for incubation [Set C**] 1.5ml tubes for supernatant [**Set D**] Boat for 85% ethanol SeraMag Bead Mixture Molecular-grade water Filtered tips

- Ensure that SeraMag bead mixture is mixed well before use; leave mixture at room temperature prior to incubation with DNA
- Pool RELIG products by assigned index (one tube per index), taking out 12µl per sample into **Set C** (DNA has already been normalised in **Section 5.1**)
- Store remaining RELIG products in freezer (in the event of future re-pooling and resequencing of samples)
- Mix DNA with desired ratio of SeraMag bead mixture; ratio is dependent on the targeted fragment size (e.g. 1.2× to retain ≥ 300bp; **Set C**)
- Leave mixture to incubate at room temperature for 15–45 mins
- Place **Set C** on magnetic rack until the beads are fully separated from the supernatant
- Draw up as much supernatant as possible without drawing up the beads (beads contain desired DNA); transfer the discarded supernatant into **Set D**
- While still on the magnetic rack, wash magnetic beads twice with 1ml 85% ethanol [**Set C**]; 85% ethanol must be freshly prepared for each clean-up
- Vacuum dry the beads for ~5 mins
- Resuspend magnetic beads in ~35µl of molecular-grade water [**Set C**]
- OPTIONAL: run the cleaned DNA and the discarded supernatant on a thick 2% agarose gel (2g agarose powder:100µl TAE buffer) to confirm purification success:
    - Load 1–1.5µl of purified DNA product side-by-side with 15–20µl of supernatant to see size ranges, selected for and discarded, respectively
    - Size ranges, as seen on the agarose gel, should not overlap
    - If desired size fragment has been discarded, top up the supernatant with more SeraMag bead mixture and repeat process to recover smaller size fragments

5.4. PippinPrep Size Selection (with 2% dye-free cassette, targeting 100–600bp)

For size selection, factor in desired size fragment (~400bp) along with adapter lengths (76bp) [2–3 hrs]

**To prepare:
0.6ml tubes for 1st elute [**Set E**]
0.6ml tubes for 2nd elute [**Set F**]
0.6ml tubes for 3rd elute [**Set G**]
Filtered tips
Loading dye
0.1% Tween20 in Qiagen Elution Buffer

- Mix **Set C** with provided loading dye, 30µl DNA:10µl loading dye
- DNA drawn for the PippinPrep size selection should be free of magnetic beads; place tubes on magnetic rack prior to withdrawal of DNA template
- The gel cassette comprises 5 lanes; 1 lane for ladder and 4 lanes for each index; load as much of sample mixed with loading dye in each well
- For PippinPrep settings, always use the "*tight*" option for a narrow size selection range
- Note down the (i) run time and (ii) elution time for future reference; the times vary from cassette to cassette:
    - E.g. To target 380bp (range of 348–412bp), the run time was 1:02:18 hrs and elution time was 3:21 mins
    - E.g. To target 415bp (range of 381–449bp), the run time was 58:18 mins and elution time was 3:31 mins
- At the eluting step, withdraw the 1st elute from each lane into respective tubes [**Set E**]
- Wash each well with buffer and transfer it into respective **Set E** tubes
- Top up the empty wells with buffer, let the protocol run for 1 min and withdraw 2nd elute [**Set F**]
- Washed with buffer again and transfer it again to **Set F**
- Repeat the above two points to get the 3rd elute [**Set G**]

5.5. SPRI Bead Purification

Clean-up of size-selected products [1 hr – 1hr 30 mins]

**To prepare:
1.5ml tubes for incubation [**Set H**]
1.5ml tubes for supernatant [**Set I**]
Boat for 85% ethanol
SeraMag Bead Mixture

Molecular-grade water Filtered tips

- Ensure that SeraMag bead mixture is mixed well before use; leave mixture at room temperature prior to incubation with DNA
- Repeat steps listed in **Section (3)**
- Mix DNA from **Set E** with desired ratio of SeraMag bead mixture [**Set H**]
- Leave mixture to incubate at room temperature for 15–45 mins
- Place **Set H** on magnetic rack until the beads are fully separated from the supernatant
- Draw up as much supernatant as possible without drawing up the beads (beads contain desired DNA); transfer the discarded supernatant into **Set I**
- While still on the magnetic rack, wash magnetic beads twice with 1ml freshly-prepared 85% ethanol [**Set H**]
- Vacuum dry the beads for ~5 mins
- Resuspend magnetic beads in ~60µl of molecular-grade water [**Set H**]
- OPTIONAL: run the cleaned DNA and the discarded supernatant on a thick 2% agarose gel to confirm purification success; if the ratio of SeraMag beads:DNA ratio used was the same as in the previous step, the size selection should work fine and there is no need to run a gel to check size selection success

5.6. PCR Enrichment of Size-selected Fragments

Triplicate enrichment of size-selected fragments per index [1 hr – 1hr 30 mins]

To prepare: Strip tubes for PCR [Set J**] 0.6ml tubes for master mix Filtered tips

- Reaction mixture as follows:

Reagent	*Final Concentration*	*1×*
5× Q5 Reaction Buffer	1×	5µl
10mM total dNTPs	800µM total dNTPs	2µl
5µM PCR1 primer	0.5µM	2.5µl
5µM PCR2 primer	0.5µM	2.5µl
Q5 HF DNA Polymerase	0.02U/µl	0.25µl
DNA Template	-	12.75µl
Total	-	25µl

- PCR1 primer is standard for all libraries; PCR2 primer is unique to each index

- Selection of PCR2 primers must be carefully done with base diversity in mind; ensure that base diversity is evenly distributed across the libraries as well as the sequencing spikes
- DNA template drawn for the PCR reactions should be free of magnetic beads; place tubes on magnetic rack prior to withdrawal of DNA template
- Protocol for RAD PCD:
  - 98ºC for 60 secs
  - 98ºC for 10 secs }
  - 68ºC for 30 secs } 8–12 cycles
  - 72ºC for 30 secs }
  - 72ºC for 10 mins
  - Hold at 15ºC
- Number of cycles is contingent on the amount of starting DNA; user must strike a balance between sufficient starting DNA material for library submission and avoiding PCR amplification bias with too many PCR cycles
- Load 1–1.5µl of PCR amplicons on 1.5–2% agarose gel to confirm amplification success and that fragment sizes across PCR replicates and libraries are similar

5.7. <u>SPRI Bead Purification</u>

Final clean-up of RAD PCR products [1 hr – 1hr 30 mins]

To prepare: 1.5ml tubes for incubation [Set K**] 1.5ml tubes for supernatant [**Set L**] Boat for 85% ethanol SeraMag Bead Mixture Molecular-grade water Filtered tips

- Repeat steps listed in **Section 5.3 and 5.5**
- Pool the replicates by library in **Set J** into respective tubes [**Set K**]
- Mix DNA with desired ratio of SeraMag bead mixture [**Set K**]
- Leave mixture to incubate at room temperature for 15–45 mins
- Place **Set K** on magnetic rack until the beads are fully separated from the supernatant
- Draw up as much supernatant as possible without drawing up the beads (beads contain desired DNA); transfer the discarded supernatant into **Set L**
- While still on the magnetic rack, wash magnetic beads twice with 1ml freshly-prepared 85% ethanol [**Set K**]
- Vacuum dry the beads for ~5 mins
- Resuspend magnetic beads in ~30µl of molecular-grade water [**Set K**]

- OPTIONAL: run the cleaned DNA and the discarded supernatant on a thick 2% agarose gel to confirm purification success

5.8. Qubit and Pooling for Library Submission

DNA quantification of RAD libraries, and final pooling across libraries depending on desired read coverage

To prepare: 1.5ml tube for pooling [Set M**] Filtered tips

- Perform DNA quantification of cleaned RAD libraries; can use either Qubit BR dsDNA or Qubit HS dsDNA kits depending on the initial DNA concentration, using 1–2µl per sample
- Assign number of reads per library in the sequencing lane and pool accordingly into **Set M**
- Concentrate **Set M** to ~30µl and run Qubit again to obtain final concentration of sample submitted for sequencing

6. Safety Precautions

1. Ensure that filter tips are used with matching pipettes to minimise risk of spills. Specifically, check that (a) the model number of the pipette and pipette tip box matches, (b) the pipette does not touch the filter, and (c) the pipette tip is firmly attached to the pipette before use.
2. Do not pipette repetitively, continuously for extended periods of time. Where possible, take frequent, short breaks every 15–20 minutes during long periods of pipetting.
3. Contact with glass bottle or conical flask containing hot agarose may cause blistering or skin burns to the person handling it. Use heat-insulating mitt, wear latex or nitrile gloves and lab coat; no direct contact with interior surface of microwave.
4. Contact with electrical components of microwave, gel electrophoresis and gel documentation systems can cause electric shock. Ensure equipment is properly connected through approved main plugs and properly grounded; ensure gel tank is closed with lid, and cables are properly connected to electrodes before operation.
5. GelRed is a sensitive, stable and environmentally safe fluorescent nucleic acid dye designed to replace the highly toxic ethidium bromide for staining DNA or RNA in agarose gels or polyacrylamide gels. It has been shown by independent tests that GelRed is neither cytotoxic nor mutagenic at concentrations well above working concentrations. However, follow the safety precautions set by the department: collect GelRed waste and dispose it as for cytotoxic waste (i.e. ethidium bromide waste). Wear latex or nitrile gloves and lab coat when handling the agarose gel and GelRed. Never use gloves that have come into contact with GelRed for handling other lab equipment and samples.

6. The transilluminator produces UV light. Do not look into the UV source without putting on UV-blocking eyewear. Ensure transilluminator door is closed properly before switching on UV light.
7. Contact with electrical components of thermal cycler and vacuum centrifuge can cause electric shock. Ensure equipment is properly connected through approved main plugs and properly grounded; ensure centrifuge is balanced and closed securely before operation.
8. Ethanol is volatile and flammable; liquid, vapours, fumes or mist may cause irritation. Store in approved flammable liquid storage containers, away from incompatible materials; only use in a cool, dry well-ventilated area away from sparks, flames and other sources of ignition; wear nitrile gloves and lab coat; keep chemicals away from exposed skin and eyes.

7. Supplementary Information

Google Drive: Reef Ecology Lab ddRADseq

http://drive.google.com/open?id=1i8eDzvnWaBdAXUvKnGOnBC2Dn1l5aP4z

- File contains all the oligonucleotide sequences needed for ddRADseq

8. Further Reading

Peterson, B. K., Weber, J. N., Kay, E. H., Fisher, H. S., & Hoekstra, H. E. (2012). Double digest RADseq: An inexpensive method for *de novo* SNP discovery and genotyping in model and non-model species. *PLoS ONE*, 7(5), e37135. doi:https://doi.org/10.1371/journal.pone.0037135

Tay, Y. C., Chng, M. W. P., Sew, W. W. G., Rheindt, F. E., Tun, K. P. P., & Meier, R. (2016). Beyond the Coral Triangle: High genetic diversity and near panmixia in Singapore's populations of the broadcast spawning sea star *Protoreaster nodosus*. *Royal Society Open Science*, 3(8), 160253. doi:10.1098/rsos.160253

Tay, Y. C., Ng, D. J. J., Loo, J. B., Huang, D., Cai, Y., Yeo, D. C. J., & Meier, R. (2018). Roads to isolation: Similar genomic history patterns in two species of freshwater crabs with contrasting environmental tolerances and range sizes. *Ecology and Evolution*, 8(9), 4657–4668. doi:10.1002/ece3.4017